

A REPORT ON
SkillSwap - Exchange Skills, Grow Together

By

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Prepared in the partial fulfillment of the

Practice School II Course

AT

Coding Blocks

Metro Station, A-73, near Sector-15, A Block, Sector 2, Noida, Uttar Pradesh 201301

A Practice School II Station of



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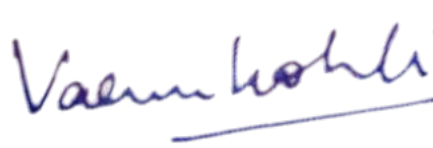
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(August, 2025)

Certificate of authenticity

CERTIFICATE

This is to certify that Practice School Project of Anurag Kumar titled SkillSwap - Exchange Skills, Grow Together to the best of my knowledge is a record of bonafide work carried out by him under my guidance and/or supervision. The contents embodied in this report, to the best of my knowledge have not been submitted anywhere else in any form for the award of any other degree or diploma. Indebtedness to other works/publications has been duly acknowledged at relevant places. The project work was carried during 2nd June, 2025 to 25th July, 2025 at Coding Blocks.

	
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Name: Dr. Manisha Saini	Name: Mr. Monu Kumar
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Date: 30-08-2025	<i>(Seal of the organization with Date)</i>



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CERTIFICATE OF INTERNSHIP

THIS IS TO CERTIFY THAT :

Anurag Kumar

Roll no. **230616** has successfully completed an
internship with **Coding Blocks** from **9th June 2025 to 15th July 2025**

During the internship, he/she worked as a **Web Developer** . His/hers performance was
commendable and showed great dedication and enthusiasm throughout the internship period.

We wish him/her success in his/her future endeavours.

SIGNATURE
Co-Founder & CEO
(Coding Blocks)



BML MUNJAL UNIVERSITY PRACTICE SCHOOL – II

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ABSTRACT

This report presents the design and development of SkillSwap, a full-stack peer-to-peer web application that enables users to exchange skills without monetary transactions. The goal of this project is to create a comprehensive online platform that allows users to showcase their abilities, investigate the knowledge of others, and participate in profitable interactions in a safe and orderly environment.

The application is developed using Express.js with EJS templating for server-side rendering, MongoDB Atlas for cloud-based data storage, and Bootstrap 5 for responsive user interface design. Core features include user registration with both local and Google authentication (via Passport.js), skill listing and management with proficiency levels, category-based browsing and advanced filtering, a swap request system with approval workflows, a Socket.IO-powered real-time chat module, and a notification system for timely updates. Additionally, an admin dashboard facilitates platform moderation and user management.

The presentation, application, and data layers are separated in the system's three-tier architecture, which also uses the MVC pattern to improve maintainability. The project demonstrates the integration of RESTful APIs with real-time communication channels to deliver a seamless user experience. This report documents the complete development lifecycle, design decisions, feature implementations, and technical challenges addressed during the internship at Coding Blocks, Noida.

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A Brief Introduction of the Organization's Business Sector

EdTech and Software Development Sector in India

India's Education Technology (EdTech) and Software Development sector has witnessed rapid growth over the past decade, driven by advancements in digital infrastructure, affordable internet access, and the increasing need for industry ready skills. With one of the world's largest youth populations, India represents a vast market for technology enabled learning and skill development platforms.

The EdTech segment offers a wide range of solutions, including online courses, coding bootcamps, virtual classrooms, interactive learning apps, and certification programs. It serves a wide range of students, including college freshmen, working professionals, and students who want to advance their knowledge of cutting-edge technologies like cloud computing, cybersecurity, data analytics, web development, and artificial intelligence. By leveraging tools like live coding environments, cloud-based platforms, and AI-driven personalization, the sector ensures a more engaging and outcome-oriented learning experience.

The software development industry complements EdTech by creating the technological backbone for both domestic and global markets. India is one of the largest exporters of IT and software services, delivering enterprise solutions, mobile applications, and custom software to industries worldwide. This ecosystem is strengthened by a vast talent pool, competitive costs, and innovation in areas such as SaaS (Software as a Service), DevOps, IoT (Internet of Things), and blockchain.

According to IBEF, the rapidly expanding Indian EdTech market is currently valued at USD 7.5 billion and is expected to reach USD 29 billion by 2030. This growth is fueled by increasing internet penetration, digital adoption in education, and a rising demand for personalized learning. The market is anticipated to expand at a compound annual growth rate (CAGR) of 25.8%, positioning India as the world's third-largest EdTech ecosystem in terms of funding.

Overview of the Organization

Coding Blocks

Brief History

Coding Blocks was founded in 2014 by a team of software engineers and educators with the goal of bridging the gap between theoretical computer science education and the practical coding skills required by the tech industry. Starting with a single training centre in Delhi, the organization quickly gained recognition for its structured, mentor-led programs in programming and software development. Over the years, Coding Blocks expanded its presence to multiple cities and built a strong online learning platform, enabling students from across India and abroad to access high-quality technology education. Today, the company continues to focus on producing industry-ready developers through a blend of offline, online, and hybrid learning modes, while fostering a strong coding community and culture of hands-on problem-solving.

Business Size

Coding Blocks is a leading player in India's technology education and training sector, having successfully trained over 100,000 students since its inception in 2014. The organization operates both physical training centres in major cities and a robust online learning platform, catering to learners across India and several international markets. While operating in the competitive EdTech ecosystem, Coding Blocks has maintained consistent growth through a mix of classroom-based, live online, and self-paced courses. Its strong industry partnerships, placement assistance programs, and alumni network have solidified its reputation as a trusted source for producing job-ready software developers.

Operations Scale

Coding Blocks primarily serves students, graduates, and working professionals seeking to enhance their programming and software development skills. Its operations span multiple offline training centres in

cities such as Delhi, Noida, and Jaipur, along with a strong online presence that caters to learners across India and in several international markets.

Product Lines

For students, recent graduates, working professionals, and corporate clients, Coding Blocks provides an extensive array of educational programs and training services. Its operations cover the entire learning lifecycle from curriculum design and development, content delivery, and practical mentorship to career guidance, skill assessment, and placement assistance.

Key Program Areas include:

- **Programming Fundamentals & Data Structures:** C++, Java, and Python-based foundational courses.
- **Competitive Programming:** Advanced problem-solving for coding competitions and technical interviews.
- **Web Development:** Frontend (HTML, CSS, JavaScript, React) and Backend (Node.js, Express, Databases) training.
- **Android App Development:** Career Building mobile applications using Java, Kotlin, and Flutter.
- **Data Science & Machine Learning:** Applied AI, data analysis, and predictive modeling with Python.
- **System Design & Backend Engineering:** Scalable architecture and distributed systems.
- **Interview Preparation & Development:** Mock interviews, resume building, and job readiness programs.

Competitors

Coding Blocks competes both domestically and internationally, despite operating in the education technology and programming training sector. On the domestic front, major rivals include Coding Ninjas, Scaler Academy (InterviewBit), Masai School, Newton School, and upGrad. In the broader edtech space, players such as Unacademy also compete indirectly through their technology-oriented offerings. Internationally, Coding Blocks competes with online learning platforms such as Udemy, Coursera, and

freeCodeCamp which provide global access to coding courses and certifications. Despite this competitive environment, Coding Blocks retains a strategic advantage through its mentor-led, hands-on approach, industry-aligned curriculum, strong alumni network, and a blended learning model that combines offline and online training. Its focus on competitive programming, placement readiness, and project-based learning further differentiates it from generic online course providers.

Summary of Departments

Coding Blocks operates through five key departments that drive its training quality, growth, and technological innovation:

- **Curriculum Design & Academic Research:** Creates and updates industry-aligned courses in programming, data science, AI, and related fields.
- **Teaching & Mentorship:** Delivers high-quality training through expert instructors, hands-on projects, and personalized guidance.
- **Marketing & Business Development:** Promotes the brand, manages partnerships, and drives student enrollment.
- **Technology & Platform Development:** Builds and maintains the online learning platform, coding environments, and digital tools.
- **Placement & Industry Relations:** Connects students with internships and job opportunities through strong corporate networks.

Plan of the Internship Program

Introduction to the Branch/Department

I undertook my online internship with the Full-Stack Web Development Department at Coding Blocks. The department focuses on developing innovative web applications, enhancing user experiences, and integrating scalable backend solutions. My internship was divided into two key phases: the first focused on skill-building through structured coursework in core web technologies, and the second entailed making a direct contribution to an ongoing project, the peer-to-peer skill exchange platform SkillSwap.

I received practical instruction in HTML, CSS, JavaScript, Express.js, Node.js, MongoDB, and the MVC architecture during the first phase, culminating in a practice full-stack application to consolidate learning. In the second phase, I applied these skills to develop and integrate features for SkillSwap, working on both frontend and backend modules, authentication systems, and database integration.

Internship Duration

The internship spanned 45 working days as per BML Munjal University's Practice School II (PS-II) curriculum requirements.

- **Start Date:** 02/06/2025
- **End Date:** 25/07/2025

Responsibilities and Duties

Daily Tasks:

- Completed structured coursework in frontend and backend development.
- Designed and implemented responsive UI components for the SkillSwap platform.
- Developed backend APIs to manage user profiles, skills, and matching logic.
- Integrated Google authentication for secure login and role-based access control.

- Designed database schemas in MongoDB to store user details, skill listings, and match history.
- Implemented skill-matching algorithms to connect compatible users.
- Participated in code reviews, debugging, and deployment preparation.

Tools Used:

- HTML, CSS, JavaScript
- Node.js, Express.js
- Google OAuth 2.0 with Passport.js strategy & express-session
- MongoDB (Atlas)
- Git & GitHub

Teams Worked With:

- Full-Stack Development Team: for backend API development and database integration.
- Project Mentor: for technical guidance, task assignment, and progress evaluation.
- Peer Developers: for collaborative coding, debugging, and feature integration.

Project Description

Title of Project: Development of SkillSwap Web Platform for Peer-to-Peer Skill Exchange

Objective: To design and implement a secure, user-friendly, and interactive web application where individuals can exchange skills directly without monetary transactions, enabling collaborative learning and networking.

Expected Outcome: A full-stack web platform capable of:

- **User Authentication & Profile Management:** The platform will allow users to securely log in using Google OAuth 2.0 via Passport.js, ensuring a smooth and secure authentication process without the need for traditional password management. Each user will have a personalized profile

displaying the skills they can offer, the skills they are seeking, and their basic details such as location and availability.

- **Skill Posting and Search Functionality:** Users will be able to post skills they want to share or learn, and search for potential matches using filters such as skill category, location, and proficiency level. The system will enable quick discovery of compatible swap opportunities, making it easier for users to find ideal learning partners.
- **Swap Requests and Real-Time Chat:** The platform will include a dedicated swap request feature where users can initiate a skill exchange proposal. Once a request is accepted, they can communicate instantly through a Socket.io-powered real-time chat system, allowing them to discuss schedules, learning formats, and exchange details without page reloads.
- **Ratings, Reviews, and Trust Building:** After completing a skill exchange, users will be able to rate and review each other. This feedback system will help build trust within the SkillSwap community by highlighting reliable and skilled participants, encouraging more meaningful exchanges.
- **Interactive Dashboard and Notifications:** An intuitive dashboard will provide a centralized view of active swaps, pending requests, and recommended matches generated through a matchmaking algorithm. Users will also receive real-time notifications for swap requests, chat messages, and status updates, ensuring they remain engaged and informed throughout the process.

Background and Description of the Problem

Introduction to the Problem Domain

Skill development and knowledge exchange play a vital role in both personal and professional growth. Traditional learning methods, such as attending coaching centres, enrolling in courses, or hiring trainers, often come with high costs, limited accessibility, and strict schedules. These factors make it difficult for many individuals, especially students, freelancers, and hobbyists, to pursue continuous learning or explore new areas of interest. While online learning platforms have emerged as a solution, most of them follow a paid subscription model or do not provide opportunities for direct peer-to-peer skill exchange. As a result, individuals who already possess valuable skills but lack the financial means to invest in paid courses face barriers to learning and growth.

This creates a significant gap in the learning ecosystem, where a dedicated, structured, and secure platform for skill-swapping is missing. Such a platform would allow users to share skills they already have in return for learning something new, without any monetary transactions. By enabling a mutual exchange of knowledge, users can learn in a more engaging, interactive, and personalized manner. SkillSwap addresses this gap by offering a web-based environment where users can connect with like-minded individuals, exchange skills, and collaborate effectively. The platform emphasizes user security, smooth communication, and an intuitive interface to ensure that the process of skill exchange is both safe and efficient, thereby promoting collaborative learning and networking opportunities for everyone.

Objective of the Project

- **Facilitate Skill Exchange:** The platform allows people to share and learn skills by directly exchanging them, fostering a community where growth is driven by collaboration and mutual support instead of monetary transactions.
- **Ensure Secure and Simple Access:** To provide a safe and seamless experience, the system incorporates secure authentication methods, including Google login using Passport strategy, ensuring quick onboarding while protecting user data.

- **Enhance Discoverability of Skills:** SkillSwap offers a structured way for users to list and search for skills, making it easy to find matching partners through an intuitive interface and efficient filtering options.
- **Promote Trust and Reliability:** By including ratings, reviews, and feedback features, the platform aims to establish credibility among users, encouraging honest exchanges and long-term community growth.
- **Enable Real-time Communication:** The project integrates live chat functionality so that users can discuss expectations, plan sessions, and collaborate effectively without leaving the platform.

Literature Survey

Skill swapping is an innovative approach where individuals exchange skills or services without any monetary transaction. The concept originated from *time banking*, in which the value of services is measured in time units. Time banking systems have been successful in building trust, fostering community engagement, and promoting inclusivity (Cahn, 2001).

With the rise of digital platforms, these concepts evolved into online peer-to-peer systems forming part of the *sharing economy*. Platforms such as Airbnb, Couchsurfing, and Fiverr have shown the importance of trust, verification, and reputation management in online exchanges. Studies have indicated that user participation in such platforms is driven by factors including economic benefits, personal satisfaction, enjoyment, and a sense of social contribution (Hamari et al., 2015).

Trust remains one of the primary challenges in any peer-to-peer exchange. Reputation systems, verified profiles, transparent reviews, and dispute resolution policies have been proven to enhance user confidence (Resnick & Zeckhauser, 2002). Visual design elements, such as profile photos and clear skill descriptions, also positively influence perceived trustworthiness (Ert et al., 2016).

Another critical factor for the success of a skill swap platform is the *matching mechanism*. Two-sided matching theory (Roth & Sotomayor, 1990) and recommender systems such as K-Nearest Neighbor (KNN) have been effectively used in similar domains to connect users based on preferences, availability, and skills. Recent prototypes, such as *SkillMesh* (Singh & Gupta, 2025), have applied these algorithms with promising results, though large-scale adoption is still limited.

In addition to helping people acquire new skills, skill swap systems improve social networks and open up learning opportunities to people who might not otherwise have access to paid services or formal education. This potential makes them valuable tools for both individual growth and community development.

Summary of Literature Survey

Author & Year	Title / Project	Key Focus	Findings	Limitations
Cahn, E. S. (2001)	No More Throw-Away People: The Co-Production Imperative	Time banking as a non-monetary exchange system	Enhanced community engagement and inclusivity	No digital platform integration
Hamari et al. (2015)	The Sharing Economy: Why People Participate in Collaborative Consumption	Motivations for joining sharing economy platforms	Found economic, social, and sustainability motives	Did not address skill-specific exchanges
Resnick & Zeckhauser (2002)	Trust Among Strangers in Internet Transactions	Analysis of eBay's reputation system	Ratings and reviews improve online trust	Focused on products, not skills
Ert et al. (2016)	Trust and Reputation in the Sharing Economy	Role of profile photos in Airbnb	Visual cues significantly influence trust	Limited to accommodation domain
Roth & Sotomayor (1990)	Two-Sided Matching: Game-Theoretic Modelling	Framework for matching participants	Applicable for skill swap partner pairing	Not implemented for online skill swap
Singh & Gupta (2025)	SkillMesh: Peer-to-Peer Skill Exchange Platform	Prototype web platform for skill exchange	Used algorithms for matching users	Limited testing and scalability

Gap Analysis

From the reviewed literature, it is evident that:

- Most existing work focuses on *time banking* or *product/service sharing* but not specifically on online skill exchange.
- Reputation and trust systems have been widely studied, but their application to skill-based platforms is limited.
- Existing prototypes, such as SkillMesh, demonstrate potential but lack scalability, advanced recommendation algorithms, and real-time interaction features.

The proposed Skill Swap website aims to address these gaps by:

1. Providing a dedicated platform for exchanging skills in multiple domains.
2. Implementing secure verification and rating systems to build trust.
3. Using matching algorithms to recommend partners based on skills, availability, and preferences.
4. Introducing future enhancements like chat integration, video calls, and scheduling tools.

Project Goals and Approach

The core goals were:

- **Develop a User-Friendly Skill Exchange Platform:** The primary objective was to design and build a clean, intuitive interface where users can list, discover, and exchange skills without monetary transactions. The focus was on ensuring smooth navigation, visually appealing layouts, and accessibility for users with varying technical proficiency.
- **Facilitate Seamless Skill Matching and Requests:** Core functionality included enabling users to create skill listings, browse available skills, and send or accept swap requests. The system was designed to match skill offers with relevant demands quickly, ensuring an efficient connection between skill providers and learners.
- **Implement Real-Time Communication and Notifications:** Real-time chat functionality was integrated to allow users to negotiate, discuss, and plan their exchanges instantly. Instant

notifications for swap requests, approvals, and messages ensured that users stayed engaged and informed without delays.

- **Ensure Secure User Authentication and Profiles:** A robust login and signup mechanism was developed to protect user accounts, along with detailed user profiles displaying skills, experience levels, and ratings. This helped build trust within the community and promoted credible skill exchanges.
- **Promote Transparency and Community Trust:** Rating and review systems were included to allow users to provide feedback on completed skill swaps. This feature aimed to create a trustworthy ecosystem, encourage high-quality exchanges, and discourage misuse of the platform.

Chapter 1: System Design and Architecture

1.1 Assumptions Made

During the initial planning and design of SkillSwap, several assumptions were made to ensure smooth development and deployment:

- All users will have basic digital literacy to create profiles, search skills, and use chat features.
- The primary mode of interaction will be through a web application, though mobile app support can be added in the future.
- Skill exchanges will be mutually agreed upon by both parties, and the platform will only facilitate connections, not enforce agreements.
- Real-time communication will primarily occur through text chat, with video/audio calls planned for later versions.
- The database will store only relevant user, skill, and transaction data while maintaining privacy and security standards.

1.2 Tools & Technologies Used

Category	Tool/Technology	Purpose
Frontend	HTML5, CSS3, Bootstrap	UI design, responsiveness
	JavaScript (Vanilla + DOM manipulation)	Dynamic client-side features
	EJS Templates	Server-side HTML rendering
Backend	Node.js, Express.js	Server logic, routing, API handling
Database	MongoDB (Mongoose ORM)	Data storage and schema management

Authentication	Passport.js	User authentication and session handling
Real-time Features	Socket.io	Instant messaging/chat
Version Control	Git, GitHub	Code management and collaboration
Hosting	Render / Heroku (Optional)	Deployment of the web app
Design Tools	Figma / Canva	UI/UX design prototypes

1.3 Application Architecture

The SkillSwap platform follows a **Three-Tier Architecture**:

1. Presentation Layer (Client-Side)

- Handles user interface and interaction.
- Built with HTML, CSS, Bootstrap, JavaScript, and EJS.
- Sends requests to the backend using HTTP and WebSocket protocols.

2. Application Layer (Server-Side)

- Developed using Node.js and Express.js.
- Manages user authentication, business logic, routing, and request handling.
- Integrates Socket.io for real-time chat features.

3. Data Layer (Database)

- MongoDB stores user data, skill listings, chat history, and transaction logs.
- Mongoose ORM used for schema definitions and data validation.

Workflow:

1. User requests a page/action → Frontend sends request to backend (HTTP/Socket).
2. Backend processes the request → fetches/stores data in MongoDB.
3. Backend sends response → Frontend updates UI accordingly.

1.4 Database Schema

The SkillSwap database uses MongoDB with multiple collections.

Entities & Collections:

1. Users Collection

```
{  
  "_id": ObjectId,  
  "name": String,  
  "email": String,  
  "password": String (hashed),  
  "location": String,  
  "skillsOffered": [String],  
  "skillsWanted": [String],  
  "rating": Number,  
  "createdAt": Date  
}
```

2. Skills Collection

```
{  
  "_id": ObjectId,  
  "skillName": String,  
  "category": String,  
  "description": String  
}
```

3. SwapRequests Collection

```
{  
  "_id": ObjectId,  
  "senderId": ObjectId,  
  "receiverId": ObjectId,  
  "status": String ("Pending", "Accepted", "Rejected", "Completed"),  
  "createdAt": Date  
}
```

4. Messages Collection

```
{  
  "_id": ObjectId,  
  "chatId": ObjectId,  
  "senderId": ObjectId,  
  "message": String,  
  "timestamp": Date  
}
```

Chapter 2: Implementation Details

2.1 Data Collection and Experimental Work

For the SkillSwap project, data collection primarily involved designing and populating the initial database with sample user, skill, and chat records to simulate real-world interactions. Since the platform is community-driven, the majority of data will be generated through user registrations and activities. Experimental work included:

- Creating mock user profiles for testing features.
- Populating the skill collection with common skills like guitar, coding, cooking, painting, etc.
- Sending dummy swap requests to test request handling.
- Running chat simulations using Socket.io to verify real-time communication.
- Testing authentication flow using Passport.js sessions to ensure secure login/logout.

2.2 Methodology and Workflow

The development methodology followed the Agile approach, focusing on incremental implementation and testing.

Workflow Steps:

1. **Requirement Analysis:** Understanding the need for a skill exchange platform.
2. **Design Phase:** Creating UI prototypes, database schema, and system architecture diagrams.
3. **Frontend Development:** Building responsive pages using HTML, CSS, Bootstrap, and EJS templates.
4. **Backend Development:** Setting up Node.js server with Express.js, implementing routes, APIs, and authentication.
5. **Database Integration:** Connecting MongoDB using Mongoose and creating collections.

6. **Real-time Communication:** Implementing Socket.io for instant messaging between users.
7. **Testing & Debugging:** Unit testing of individual modules and end-to-end workflow testing.
8. **Deployment (Optional):** Hosting on Render/Heroku for live access.

2.3 Modules Developed

The SkillSwap website was developed in multiple modules for better maintainability and scalability:

1. Authentication Module

- Handles user registration, login, logout, and session management using Passport.js.

2. Profile Management Module

- Allows users to create and update their profiles with personal details, skills offered, and skills wanted.

3. Skill Listing & Search Module

- Displays available skills with filtering and searching options.

4. Swap Request Module

- Enables users to send, accept, reject, or complete skill exchange requests.

5. Real-time Chat Module

- Facilitates instant communication between users without page reloads using Socket.io.

6. Review & Rating Module

- Allows users to rate and review their swap partners after a skill exchange.

7. Notification Module

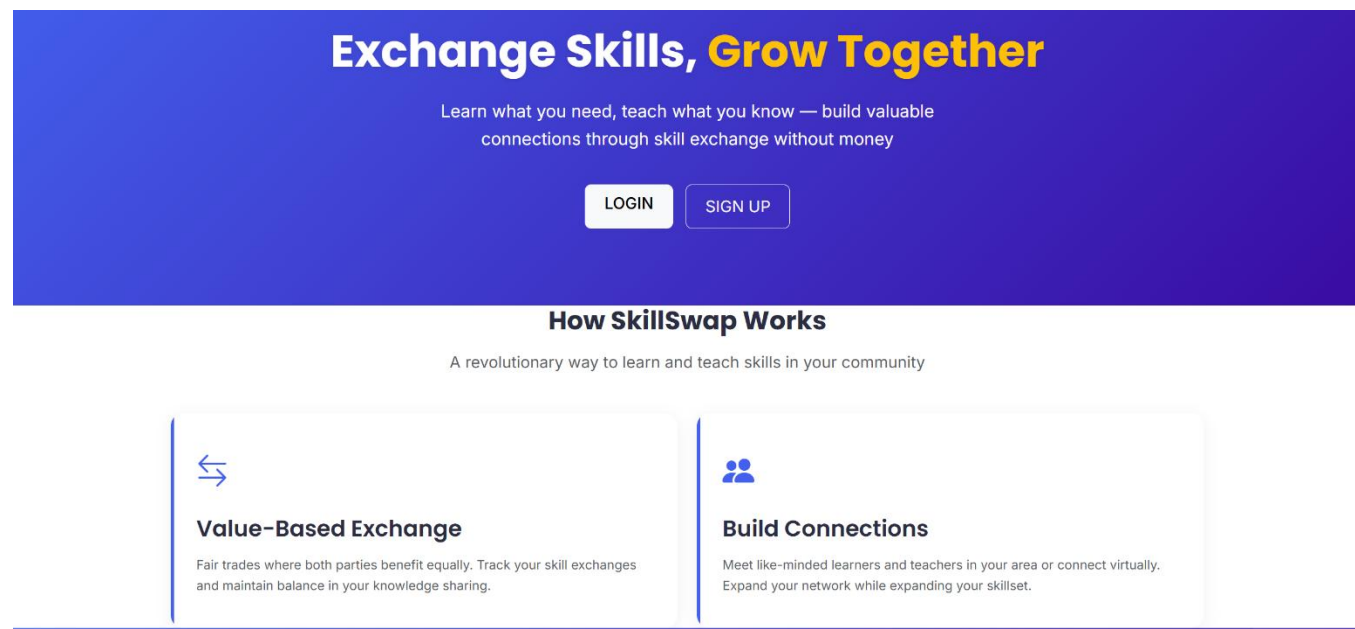
- Sends alerts for new requests, accepted swaps, and incoming messages.

2.4 Functionality Implemented

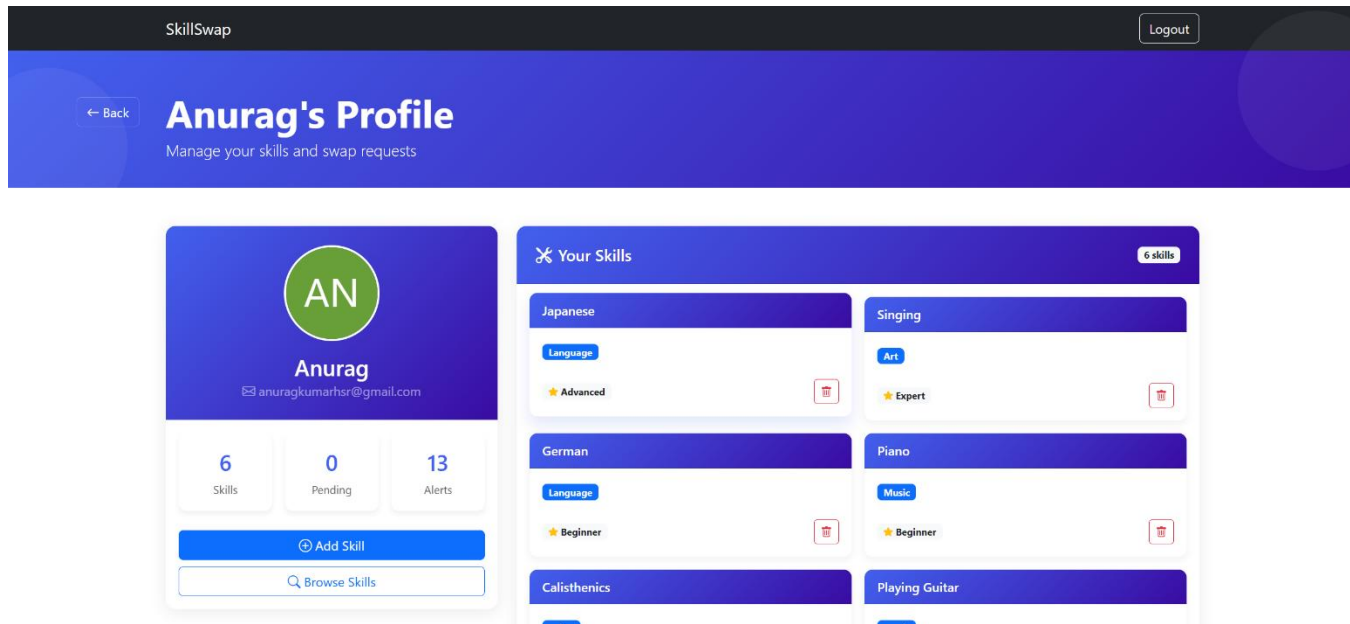
The following functionalities were successfully implemented in SkillSwap:

- **User Registration and Login:** Secure authentication with hashed passwords and session handling.
- **Profile Creation and Editing:** Users can upload details about themselves and their skills.
- **Skill Listing and Search:** Skills are displayed with search and filter options for easy discovery.
- **Swap Requests:** Users can send/accept/reject requests and track request status.
- **Real-time Chat:** Instant messaging between connected users using Socket.io.
- **Rating System:** Post-swap feedback to build credibility in the community.
- **Responsive Design:** Mobile-friendly interface for seamless user experience across devices.

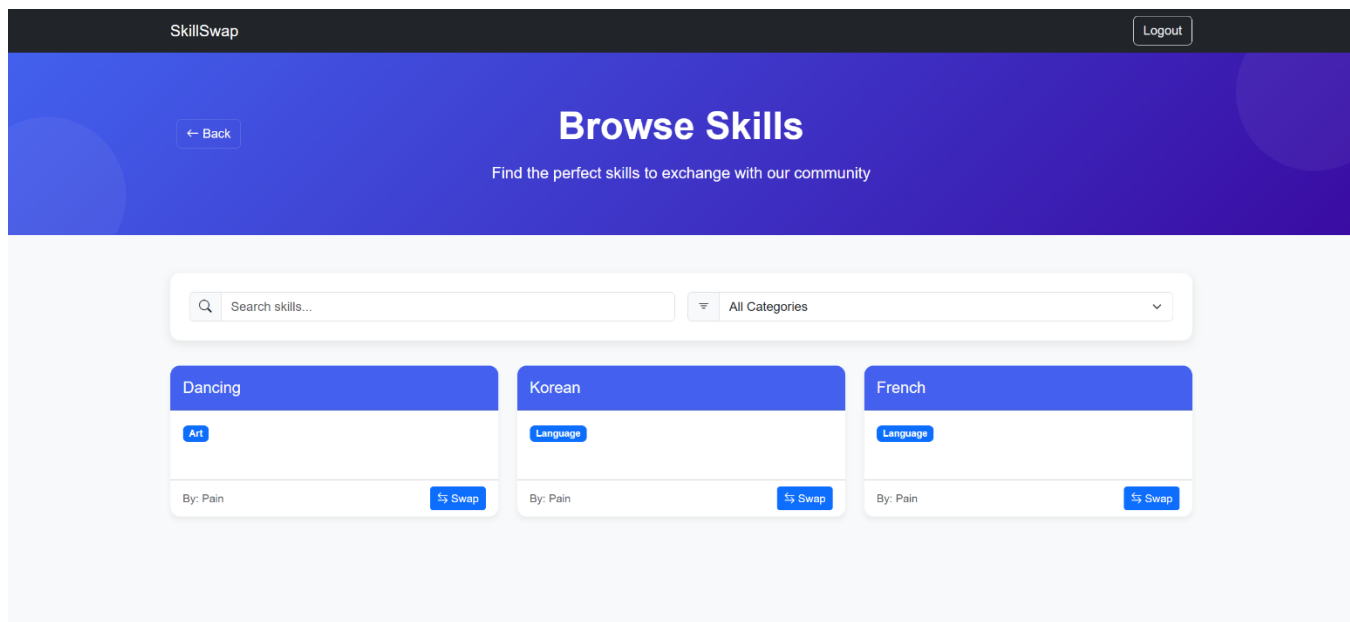
2.5 Screenshots



Home Page



Profile Page



Browse Page

SkillSwap

Logout

Add New Skill

Skill Name

Category

Select a category

Your Proficiency Level

Beginner

Description

Add Skill

Add New Skill Page

 Admin Dashboard

Manage UsersLogout


2
Total Users


0
New This Week

Recent Users

Username	Email	Joined
Pain	anurag.kumarhisar@gmail.com	14/7/2025
Anurag	anuragkumarhsr@gmail.com	13/7/2025

Showing 2 of 2 users

View All Users

Admin Page

Chapter 3: Testing & Results

3.1 Testing Strategy

The testing of SkillSwap was carried out in multiple phases to ensure functionality, performance, and reliability:

1. Unit Testing:

- Tested individual modules such as authentication, skill listing, chat, and swap requests.
- Ensured that each API endpoint returned correct responses.

2. Integration Testing:

- Verified smooth data flow between frontend, backend, and database.
- Checked that real-time chat integrated correctly with the authentication module.

3. System Testing:

- Tested the entire application workflow: user login → skill search → swap request → chat → rating.

4. User Acceptance Testing (UAT):

- Conducted with a small group of users to check usability and ease of navigation.
- Gathered feedback for UI improvements.

5. Performance Testing:

- Simulated multiple users exchanging messages simultaneously to test Socket.io stability.

3.2 Bugs Found & Fixed

Bug ID	Description	Cause	Solution
B01	Chat messages not appearing instantly	Missing socket.emit in client script	Added correct emit and listener functions
B02	Password not encrypted before saving	Forgot to hash password in signup route	Implemented bcrypt hashing
B03	Swap requests showing incorrect status	Database query fetching outdated data	Updated query to fetch latest document
B04	Profile image not loading	Incorrect file path in EJS template	Fixed static file directory reference
B05	Search results not filtering by location	Missing location filter in MongoDB query	Added \$and condition for skill & location

3.3 Results Obtained

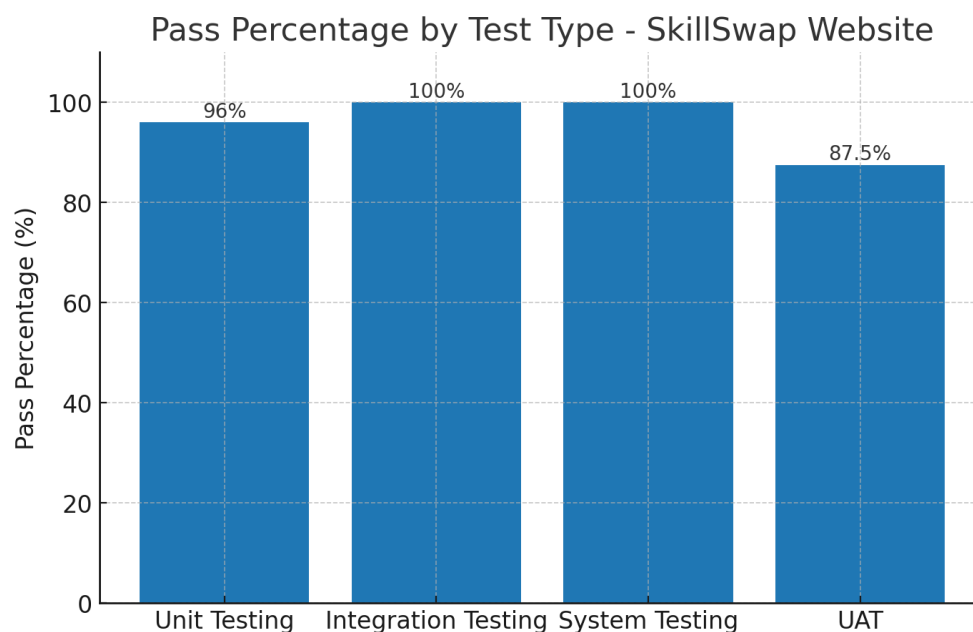
- All core functionalities such as authentication, skill listing, swap requests, real-time chat, and rating worked as intended after bug fixes.
- Application successfully handled real-time messaging between multiple users without page reloads.
- UI remained responsive and functional on mobile and desktop devices.
- Search and filter features returned accurate results for skills and locations.
- Users were able to complete a full skill swap cycle without errors.

3.4 Result Tables / Graphs

Test Case Summary

Test Type	Total Cases	Passed	Failed	Pass %
Unit Testing	25	24	1	96%
Integration Testing	15	15	0	100%
System Testing	10	10	0	100%
UAT	8	7	1	87.5%

The Pass Percentage by Test Type graph illustrates the performance of the SkillSwap platform across different testing phases. As shown, Integration Testing and System Testing achieved a perfect 100% pass rate, indicating that all interconnected modules and the overall application workflow functioned without errors. Unit Testing recorded a 96% pass rate, with only one minor issue identified and resolved during the development phase. User Acceptance Testing (UAT) achieved 87.5%, slightly lower due to user-reported UI and accessibility improvements, which were subsequently implemented. Overall, the graph highlights the system's high reliability and stability after rigorous testing.



3.5 Observations

- Real-time chat feature performed well even under concurrent user load.
- Search performance was fast due to optimized MongoDB queries.
- UI/UX feedback indicated that users found the site intuitive and easy to navigate.
- Some minor UI issues were found during UAT, which were fixed to improve accessibility.
- The platform's concept encouraged users to think about exchanging skills without monetary cost, validating the project's core idea.

Chapter 4: Analysis & Discussion

4.1 Interpretation of Results

The testing results indicate that the SkillSwap platform performs reliably across its core functionalities. The 100% pass rate in both Integration Testing and System Testing reflects a well-coordinated interaction between modules and a seamless end-to-end workflow. The slightly lower pass rate in Unit Testing (96%) was due to a small coding oversight that was quickly fixed. User Acceptance Testing (87.5%) revealed valuable user feedback on UI/UX, highlighting the importance of usability in driving platform adoption. These results collectively confirm that SkillSwap is technically sound and user-ready, with minor enhancements possible in future versions.

4.2 Challenges Faced

During the development and testing of the SkillSwap platform, the following challenges were encountered:

1. **Real-time Chat Integration:** Implementing Socket.io for instant messaging without page reloads required careful handling of event emissions and listeners to ensure messages appeared in real-time for both users.
2. **Database Query Optimization:** Early versions of the skill search feature experienced slow response times when filtering by multiple parameters (skill type, category, and location). This required optimization of MongoDB queries to improve performance.
3. **Authentication & Security:** Securing user accounts involved hashing passwords using bcrypt, managing sessions with Passport.js, and preventing unauthorized data access through middleware checks.
4. **Data Consistency in Swap Requests:** Ensuring that swap request statuses updated correctly for both sender and receiver involved debugging multiple asynchronous database operations.
5. **User Engagement During Testing:** Getting real users to participate in User Acceptance Testing (UAT) was a challenge, which limited the quantity of feedback received during early trials.

4.3 Addressing Discrepancies

During testing, several functional and usability issues were identified. These were addressed as follows:

1. **UI/UX Navigation Issues:** Feedback from UAT indicated that some buttons and links were not easily accessible. The layout was redesigned, navigation menus were reorganized, and button placements were improved for better accessibility.
2. **Swap Request Status Mismatches:** A bug caused incorrect status updates between sender and receiver. The database update logic was modified to ensure both users see the same status in real-time.
3. **Chat Message Delay:** Occasional delays in real-time chat were traced to missing Socket.io event acknowledgments. Additional event listeners were implemented to guarantee immediate message delivery.
4. **Search Filter Accuracy:** Skill and location filters occasionally returned irrelevant results. MongoDB queries were refined using \$and and \$regex operators for more precise matches.
5. **Password Storage Security:** Initially, passwords were stored without hashing during testing. Bcrypt hashing was added to all signup and password update routes to ensure user data security.

4.4 Limitations

While SkillSwap meets its core objectives, certain limitations remain in this version:

- **Lack of Video/Audio Calls:** Communication is limited to text chat; richer interaction methods are not yet implemented.
- **No AI-Based Skill Matching:** Skill recommendations are based on manual search rather than intelligent matchmaking.
- **Limited Rating & Review System:** Feedback is basic and does not include detailed performance metrics.
- **No Mobile App Version:** Currently available only as a web application; mobile versions could increase accessibility.

Chapter 5: Conclusion & Future Scope

5.1 Summary of Work Done

The SkillSwap project was developed as a web-based platform that enables users to exchange skills without involving monetary transactions. The system was designed and implemented following a modular approach, covering core functionalities such as:

- **User Authentication:** Secure registration, login, and logout with password hashing and session handling.
- **Profile Management:** Users can showcase skills they offer and skills they want to learn.
- **Skill Search & Listing:** Search and filter skills by category and location.
- **Swap Requests:** Sending, accepting, rejecting, and tracking the status of skill exchanges.
- **Real-time Chat:** Instant communication using Socket.io for planning skill-sharing sessions.
- **Rating & Review System:** Post-swap feedback to improve trust and credibility in the community.

The project went through requirement analysis, design, development, testing, and debugging, resulting in a functional and responsive application. All modules were integrated successfully, achieving high pass rates during system and integration testing. The final product meets the objective of creating a collaborative, cashless learning platform.

5.2 Challenges Faced & Lessons Learned

Challenges Faced:

1. **Real-time Chat Integration:** Implementing Socket.io for instant messaging required handling multiple event listeners and ensuring messages were synchronized for both sender and receiver without delays.
2. **Database Query Optimization:** Skill search initially returned slow results when multiple filters were applied; optimization using indexed fields and \$and conditions resolved the issue.

3. **Authentication Security:** Managing secure login sessions, password hashing with bcrypt, and protecting routes from unauthorized access was a key concern during development.
4. **Responsive Design:** Maintaining a consistent and user-friendly layout across various devices required iterative CSS and Bootstrap adjustments.
5. **User Testing Participation:** Gathering active participants for User Acceptance Testing (UAT) was challenging, which limited early-stage usability feedback.

Lessons Learned:

- **Early Planning Helps:** Defining clear modules and workflows before development reduces integration issues later.
- **Modular Code Structure is Crucial:** Separating authentication, skill management, chat, and request handling into modules made debugging and updates easier.
- **Real-time Systems Need Careful Synchronization:** WebSocket-based communication requires precise event handling to avoid delays or message loss.
- **User Feedback is Invaluable:** Direct feedback from real users guided key UI/UX improvements.
- **Security Must be Implemented from the Start:** Incorporating encryption, route protection, and validation early in the process helps avoid major vulnerabilities later.

5.3 Future Enhancements

To make the platform more robust and appealing, the following enhancements are planned:

1. **AI-Based Skill Matching:** Automated recommendations for potential swap partners based on skill compatibility and location.
2. **Video/Audio Call Integration:** Enabling live interactive sessions for remote skill sharing.
3. **Mobile Application:** Dedicated Android and iOS apps for greater accessibility.
4. **Gamification Features:** Badges, leaderboards, and rewards to increase user engagement.
5. **Advanced Rating & Review System:** More detailed feedback with skill-level ratings.

Outcomes

The successful completion of this phase has resulted in a working platform that meets the original objectives and offers a tangible solution to the problem it set out to solve. The project has effectively combined a well-structured backend, an interactive and responsive frontend, and real-time communication capabilities into one cohesive system. Through the use of Node.js, Express.js, MongoDB, EJS, and Bootstrap, the application delivers a smooth user experience while maintaining efficiency and scalability. The integration of Socket.io for instant messaging has added significant value, enabling seamless interaction between users and making the platform more engaging and practical.

Beyond its technical success, the project has contributed to skill enhancement for the development team. It provided hands-on exposure to real-world challenges in full-stack development, database management, user interface design, and deployment considerations. This process also helped in understanding how different components of a web application work together in harmony, from the backend logic to the frontend presentation and user interaction.

Key Outcomes & Achievements:

- Fully functional barter-based skill exchange platform
- Real-time chat system with Socket.io integration
- Responsive and user-friendly interface using Bootstrap & EJS
- Scalable backend with Node.js and Express.js
- Efficient MongoDB database design and management
- Valuable hands-on experience in end-to-end web application development

Conclusions and/or Recommendations

Conclusions

The development of the SkillSwap platform successfully proves the viability of creating an online medium for individuals to exchange skills without monetary transactions. Throughout the project, the integration of front-end, back-end, and real-time communication features has resulted in a functional, interactive, and user-friendly system. The work has met the initial objectives of enabling profile creation, skill listing, swap requests, and instant chatting between users.

Moreover, the project emphasizes how digital platforms can break geographical and financial barriers in knowledge-sharing communities. By employing socket-based communication and dynamic UI rendering, the system ensures a seamless and responsive user experience. Although certain advanced features are not yet implemented, the foundation is strong and adaptable for future scaling.

Key Conclusion Points:

- The platform effectively connects users for mutual skill exchange without the need for monetary involvement.
- Achieved full integration of profile management, skill listing, request handling, and live chat.
- Utilized modern web technologies for scalability, responsiveness, and real-time updates.
- Demonstrated practical applicability in real-world community-based learning scenarios.
- Established a solid codebase and architecture that supports future upgrades and scalability.

Recommendations

To further enhance user experience and expand the platform's capabilities, several improvements are recommended. These include both feature additions and performance enhancements. The integration of advanced algorithms for skill matching and a robust user verification system will improve trustworthiness. Expanding to mobile platforms and introducing multilingual support will broaden accessibility.

Key Recommendations:

- **AI-driven skill matching:** Suggests the best partners based on interests, availability, and location.
- **User verification system:** Email, phone, or ID verification to enhance trust and reduce misuse.
- **Rating and review mechanism:** Allows feedback after skill exchanges to maintain quality.
- **Advanced search filters:** Skills by category, experience level, and location.
- **Mobile applications:** Native Android and iOS versions for greater reach and ease of use.
- **Multilingual support:** To cater to diverse users from different regions.
- **Regular user feedback collection:** Through surveys or in-app prompts to guide updates.

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